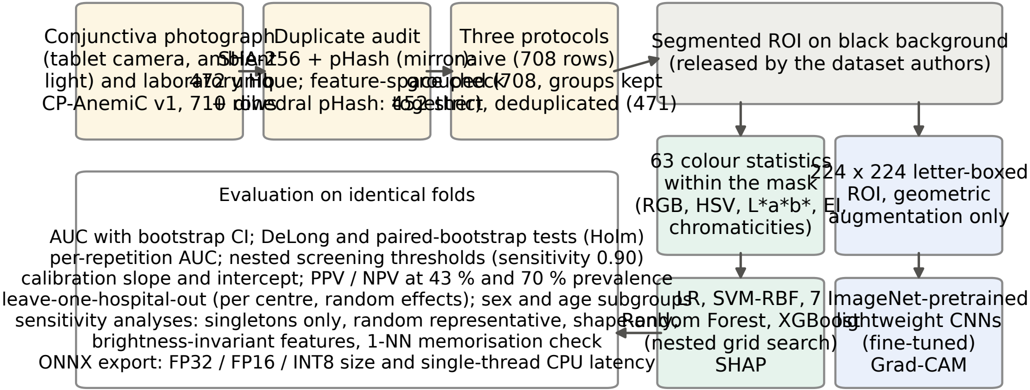
Conjunctiva photograph
(tablet camera, ambient
light) and laboratory Hb
CP-AnemiC v1, 710 rows
Duplicate audit
SHA-256 + pHash (mirror):
472 unique; feature-space check
+ dihedral pHash: 452 strict
Three protocols
naive (708 rows)
grouped (708, groups kept
together), deduplicated (471)
Segmented ROI on black background
(released by the dataset authors)
63 colour statistics
within the mask
(RGB, HSV, L*a*b*, EI,
chromaticities)
224 x 224 letter-boxed
ROI, geometric
augmentation only
LR, SVM-RBF, 7
Random Forest, XGBoost
(nested grid search)
SHAP
ImageNet-pretrained
lightweight CNNs
(fine-tuned)
Grad-CAM
Evaluation on identical folds
AUC with bootstrap CI; DeLong and paired-bootstrap tests (Holm)
per-repetition AUC; nested screening thresholds (sensitivity 0.90)
calibration slope and intercept; PPV / NPV at 43 % and 70 % prevalence
leave-one-hospital-out (per centre, random effects); sex and age subgroups
sensitivity analyses: singletons only, random representative, shape-only,
brightness-invariant features, 1-NN memorisation check
ONNX export: FP32 / FP16 / INT8 size and single-thread CPU latency